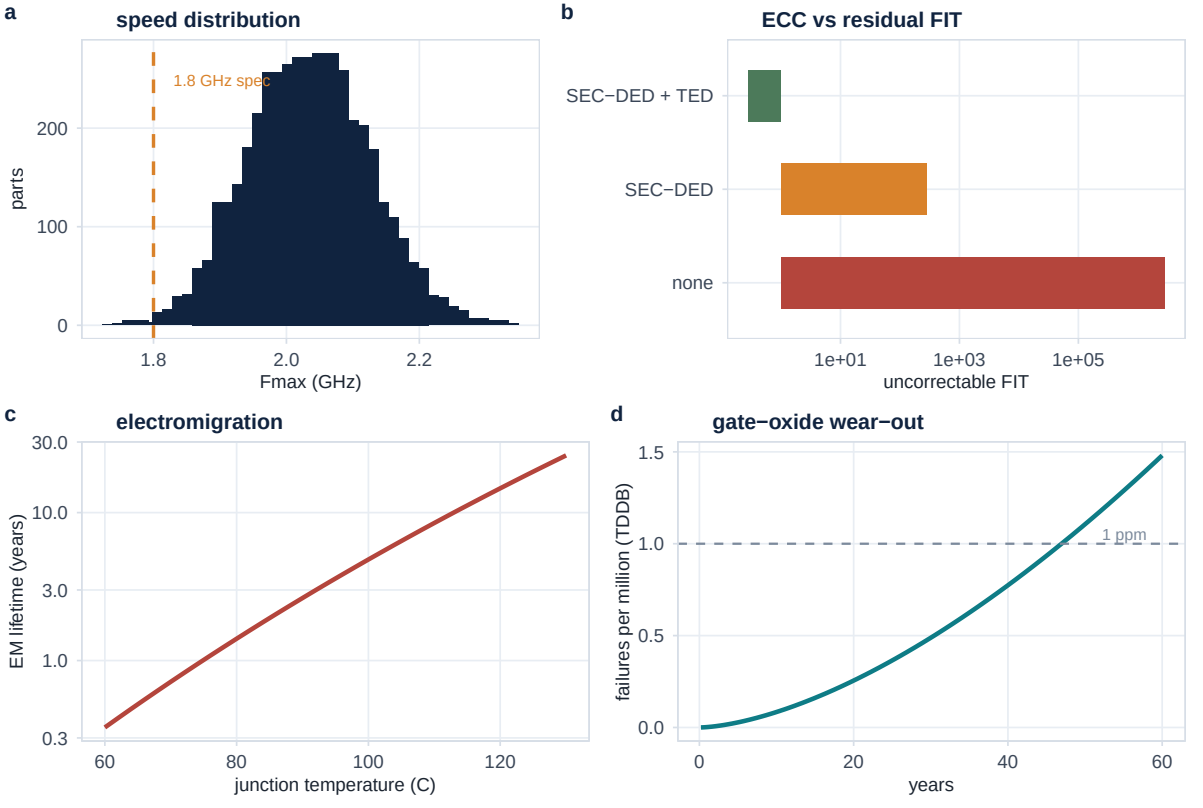
a
speed distribution
1.8 GHz spec
parts
200
100
0
1.8
2.0
2.2
Fmax (GHz)
b
ECC vs residual FIT
SEC−DED + TED
SEC−DED
none
1e+01
1e+03
1e+05
uncorrectable FIT
c
electromigration
EM lifetime (years)
30.0
10.0
3.0
1.0
0.3
60
80
100
120
junction temperature (C)
d
gate−oxide wear−out
failures per million (TDDB)
1.5
1.0
0.5
0.0
1 ppm
0
20
40
60
years